INCH-POUND

MIL-C-70571B (AR)
23 February 1993
SUPERSEDING
(See 6.6)

MILITARY SPECIFICATION

CIRCUIT BOARD, BUILT-IN-TEST BOARD NO. 1, UNIT 4: 9360949

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements and tests for the AN/VPS-2A Radar Range Computer, Built-In-Test (BIT) Board No. 1 Assembly referred to as the Unit 4BB1 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926 - Fire Control Materiel, Manufacture and Inspection, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1285

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MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9360949 - BIT Board 1 Assembly, Computer, Range
9360951 - BIT Board 1 Schematic, Computer, Range
9361338 - Test Program Set Manual
9397373 - Interface Device C
9397562 - IDC W5 Printed Wiring Board Assembly
9397563 - IDC W4 Cable Assembly
9397376 - Cable Assembly, Polarized
12950877 - Requirements for Soldering Electrical and Electronic Assemblies

SPECIAL PACKAGING INSTRUCTION

PS 9360884 - Packaging of Display Control Panel Circuit Board

(Copies of drawings, publications and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

INSTITUTE FOR INTERCONNECTING AND PACKAGING ELECTRONIC CIRCUITS

ANSI/IPC-DW-425	-	Design and End Product Requirements for Discrete Wiring Boards
EIA RS-422	-	Electrical Characteristics of Balanced Voltage Digital Interface

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10018-3308.)

2.3 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 9360884, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the Unit 4BB1 shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances.

b. Inorganic protective surface finishings.

c. Part identification and marking.

d. Workmanship.

3.2.1 ANSI/IPC-DW-425. IPC-DW-425/11 Type C1 plated through connection.

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3.3 Environmental service conditions.

3.3.1 Vibration. The Unit 4 BB1 assembly shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to the equivalent of 6000 miles of track-laying vehicle operational vibration as defined by MIL-STD-810 narrowband random-on-random data for equipment installed on the deck of the M113 armored personnel carrier.

3.3.2 Shock. The Unit 4 BB1 assembly shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ± 10%, duration 11 milliseconds ± 10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

3.3.3 Altitude operating.. The Unit 4BB1 assembly shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The Unit 4 BB1 assembly shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40^{0}F and $+155^{0}$F.

3.3.4.2 Nonoperating (storage). The Unit 4 BB1 assembly shall not suffer permanent damage when subjected to the constant temperature extremes of -65^{0}F for a period of 24 hours and $+160^{0}$F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (nonoperating). The Unit 4 BB1 assembly shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram 9360951 of the Unit 4 BB1 assembly.

3.5 Performance.

3.5.1 Functional description. The Unit 4 BB1 assembly (see figure 1) provides the ability to transmit radar dynamic range data continuously while the system is in radar mode, and to transmit static test data while in BIT mode. Received serial

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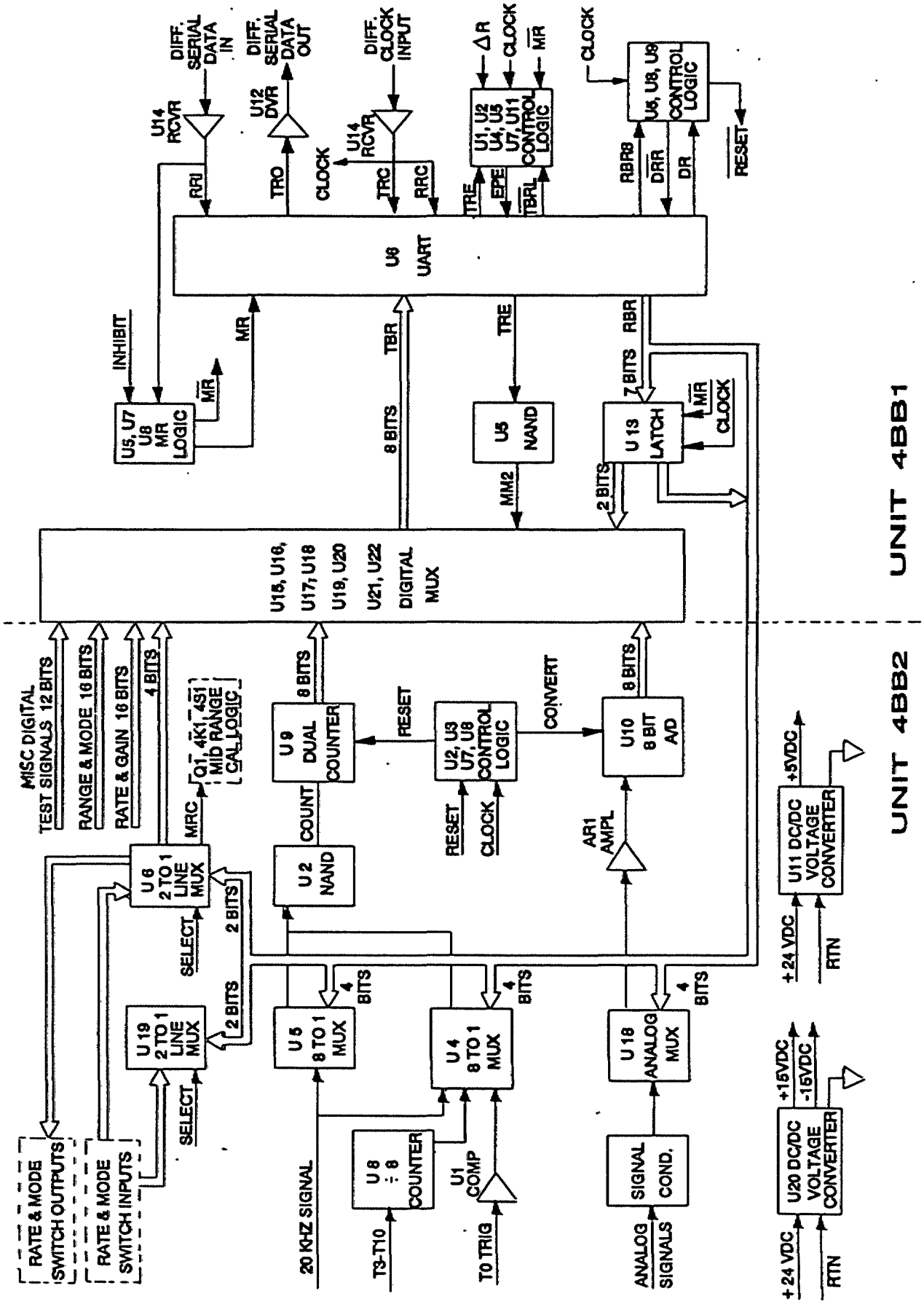


FIGURE 1. Unit 4 BIT functional block diagram.

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address data is converted to a parallel by a universal asynchronous receiver/transmitter (UART) and used to select the data channels of a 64-line to 8-line digital multiplexer. The 8 multiplester output lines are arranged to form a lower or upper byte of a 16-bit data word. The 64 multiplexer data input lines make up four 16-bit channels as follows, each with a lower and an upper byte:

a. Range and mode data channel

b. Rate and gain data channel

c. Analog voltage and digital counter data channel

d. Digital test point data channel

The lower and upper data bytes from the multiplexer are alternately parallel loaded into the UART for separate serial transmission. The 64 multiplexer input lines shall monitor data from the following radar computer line-replaceable-units (LRUs):

a. 4A1 Master Clock

b. 4A2 PRF Counter

c. 4A3 Modulator Trigger Generator

d. 4A4 Computer Timing Generator

e. 4A5 RCFD Decoder

f. 4A6 Range Counter

g. 4A7 RCFD Generator

h. 4A8 Range Register A

i. 4A9 Range Register B

j. 4A10 Search/Track and Gain Control

k. 4A11 Range Rate Counter A

l. 4A12 Range Rate Counter B

m. 4A13 Rate Scaler

n. 4A14 Rate Integrator

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o. 4A15 Phase Detector

p. 4A16 Unit 4 BIT 2

q. 4A17 Unit 4 BIT 1

r. 4A16/4A17 Ribbon Cable

s. 4A18 5-volt Filter

t. 4A19 15-volt Regulator

u. 4A20 25-volt Regulator

v. 4A21 Control Amplifier

w. 4A22 Fine/Fine Delay

x. 4A23 Universal Board

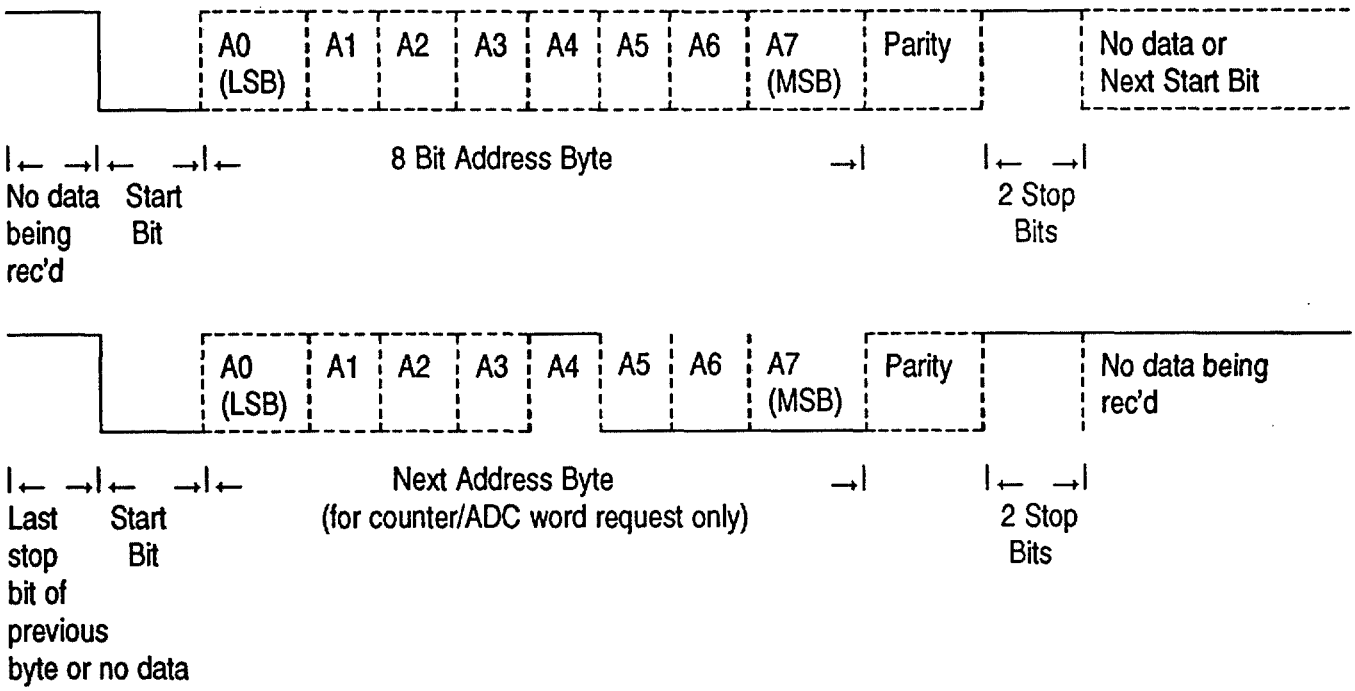
3.5.2 Power requirements. Unit 4BB1 power shall be inputted at J1-4,5 (+5 V) and J1-14,15 (+5 V RTN) with requirements as follows:

	MIN	MAX	UNITS
Input Voltage	4.75	5.25	Vdc
Input Current	---	0.5	Amperes
Input Ripple	---	150	mV p-p

3.5.3 Clock requirements. The UNIT 4 CLOCK (500 KHZ) signal applied at P1-2 (HI) and P1-3 (LO) shall be differential with RS-422 voltage levels. The clock frequency shall be 500.000 kHz ± 74 Hz. With power applied to the Unit 4BB1 per 3.5.2, the clock shall have TTL Voltage levels at CLK 0 (500 KHZ) at J1-42, except a pull-up resistor shall ensure the minimum 3.0 V CMOS threshold required by the UART.

3.5.4 Format of UART receiver input data. The address bytes inputted into the Unit 4BB1 shall be differential signals with RS-422 voltage levels applied at CPU to UNIT 4 DATA with P1-4 (HI) and P1-5 (LO). With power and clock applied to the Unit 4BB1 per 3.5.2 and 3.5.3, respectively, the serial input data shall have TTL voltage levels at J1-36 (RRI) test point, except a pull-up resistor shall ensure the minimum 3.0 V CMOS threshold required by the UART. When no address data is being received, CPU to UNIT 4 DATA and RRI shall be logic high. The input data rate shall be 31.250 kHz ± 5 Hz (one-sixteenth of the 500 kHz clock rate). The format for the input bytes shall be per Figure 2.

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NOTES:

1. The data line is always high when no address data is being received.

2. Each start bit is always low and each stop bit is always high.

3. LSB = Least Significant Bit and is the first address bit received in each byte.
   MSB = Most Significant Bit and is the last address bit received in each byte.

4. The parity bit is a "don't care" because the parity error flag output from the UART is not connected.

5. When an address byte has been received by the UART, it is output in parallel on the UART's Receiver Buffer Register lines as follows:

   Bit A0 is on RBR1
   Bit A1 is on RBR2
   ↑
   ↓
   Bit A7 is on RBR8

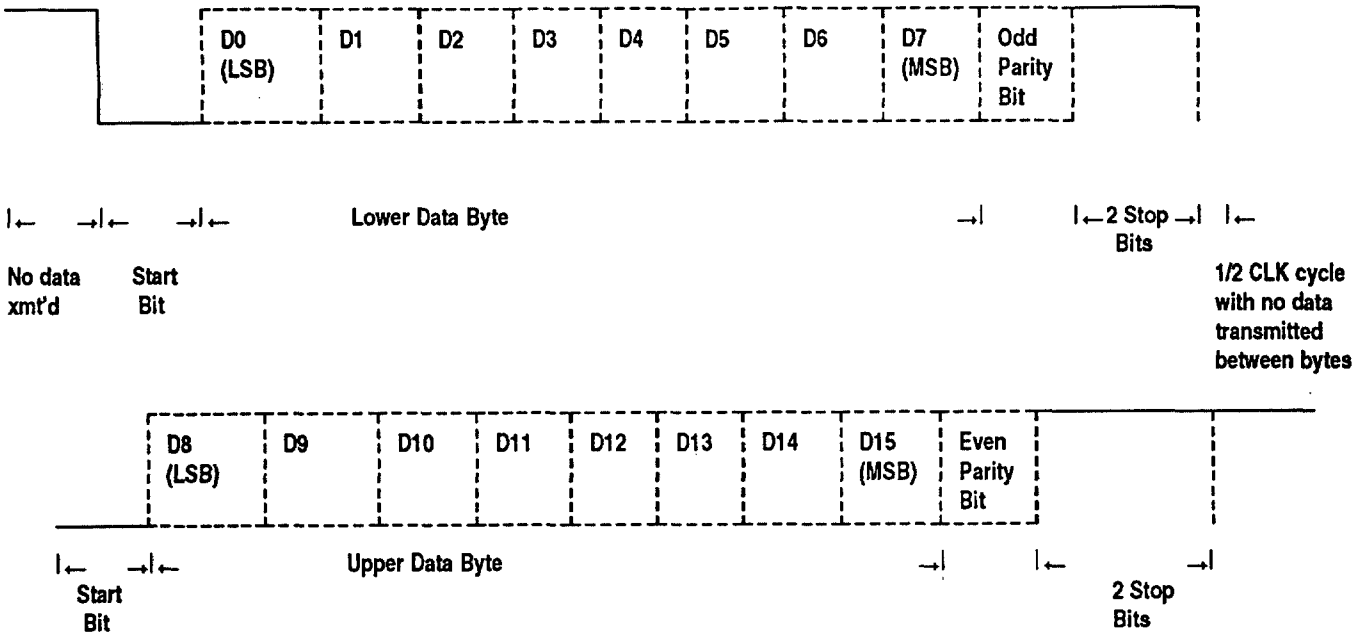
6. A0 through A3 of the second address byte received set and reset RBR1 through RBR4, and the logic high on A4 causes RST to go logic low.

FIGURE 2. Format of UART receiver input data.

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3.5.5 Format of UART transmitter output data. Each byte transmitted from the Unit 4BB1 is differential with RS-422 voltage levels and is output at UNIT 4 to CPU DATA with P1-6 (HI) and P1-7 (LO). With power and clock applied to the Unit 4BB1 per 3.5.2 and 3.5.3, respectively, each transmitted output byte shall have CMOS voltage levels at J1-34 (TRO) test point and a data frequency of 31.250 kHz ± 5 Hz. Each 16-bit data word transmitted from the Unit 4BB1 shall consist of two sequential 8-bit bytes. The first of the two bytes shall be the lower byte and contain the 8 least significant bits of the word and have odd parity. The second of the two bytes shall be the upper byte and contain the 8 most significant bits of the word and have even parity. The transmitted output data format shall be per Figure 3.

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NOTES:

1. The data line is always high when no address data is being transmitted.

2. The start bits are always low and the stop bits are always high.

3. LSB = Least Significant Bit and is the first data bit transmitted in each byte.
   MSB = Most Significant Bit and is the last data bit transmitted in each byte.

4. Data bits D0 through D15 make up one 16-bit data word when the lower and upper data bytes are reassembled. D0 is the LSB of the data word and D15 is the MSB of the data word.

5. For the odd parity bit, the UART sums data bits D0 through D7 and adds a one (makes the parity bit high) or adds a zero (makes the parity bit low) to cause the sum plus the parity bit to be an odd number.

6. For the even parity bit, the UART sums data bits D8 through D15 and adds a one (makes the parity bit high) or adds a zero (makes the parity bit low) to cause the sum plus the parity bit to be an even number.

7. The UART generates the parity bit at time of byte loading.

8. Each data bit for transmission is loaded into the UART on the UART's Transmitter Buffer Register (TBR) lines as follows:

Low Byte	High Byte
Bit D0 is on TBR1	Bit D8 is on TBR1
Bit D1 is on TBR2	Bit D9 is on TBR2
↕	↕
Bit D7 is on TBR8	Bit D15 is on TBR8

FIGURE 3. Format of UART transmitter output data.

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3.5.6 Master reset operation.

3.5.6.1 Master reset upon power-up. When power and clock are applied to the Unit 4BB1 per 3.5.2 and 3.5.3, respectively (no data input), Q1 (J1-44) of the time-out circuit shall remain low. Also upon power-up, the Master Reset (MR) line to the UART shall go high for 1.2 to 2.8 milliseconds before returning low and the MR (J1-17) line to the control circuitry shall remain low for 1.2 to 2.8 milliseconds before going and remaining high. When the MR line pulses high, the UART shall reset, as evidenced by the following:

a. DR (J1-9) shall be and remain low.

b. MM2 (J1-41) shall go high for less than 18 clock cycles then go and remain low.

c. TRO (J1-34) shall go and remain high.

The $\overline{MR}$ line low pulse shall cause the following steady state conditions:

d. $\overline{DRR}$ (J1-35) shall be high.

e. MM0 (J1-28) shall be low.

f. MM1 (J1-39) shall be low.

g. S1 (J1-24) shall be low.

h. S6 (J1-50) shall be low.

i. S4 (J1-25) shall be low.

j. S3 (J1-23) shall be low.

k. SW2 (J1-48)/TP9 shall be low.

l. Q2 (J2-9) shall be low.

m. Q3 (J2-10) shall be low.

n. EPE (J1-16) shall be low.

o. $\overline{TI}$ (J1-1) shall be high.

p. $\overline{TBRL}$ (J1-2) shall be high.

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3.5.6.2 Master reset after receiving data. With power, clock and address data applied to the Unit 4BB1 per 3.5.2, 3.5.3 and 3.5.4, respectively, every negative going data pulse edge on the RRI line shall initiate a 144.5 ± 17.5 millisecond positive delay pulse on the Q1 (J1-44) line. Q1 shall remain high while being retriggered if the previous delay has not timed out. The pulses shall be non-cumulative in that Q1 shall go low 144.5 ± 17.5 milliseconds after the last negative going data pulse edge. When Q1 goes low, MR shall go high for 0.3 to 2.0 milliseconds and MR (J1-17) shall go low for 0.3 to 2.0 milliseconds. After the reset, the Unit 4BB1 signal logic levels shall be the same as those of 3.5.6.1 a through p.

3.5.7 UART receiver circuit.

3.5.7.1 Initialization. With power and clock applied per 3.5.2 and 3.5.3, respectively, the Unit 4BB1 shall initialize (reset per 3.5.6) for receiving data.

3.5.7.2 UART receiver operation. (Refer to Figure 4.) Clocking differential serial data per 3.5.4 into the Unit 4BB1 shall cause DR to go high, signifying that new received data (bits A0 through A7) are ready in parallel form on Receiver Buffer Register lines RBR1 through RBR8, respectively. The rising edge of the first clock pulse after DR goes high shall cause DRR to go low, resetting DR low for receiving the next address byte. DRR shall return high on the following rising clock edge.

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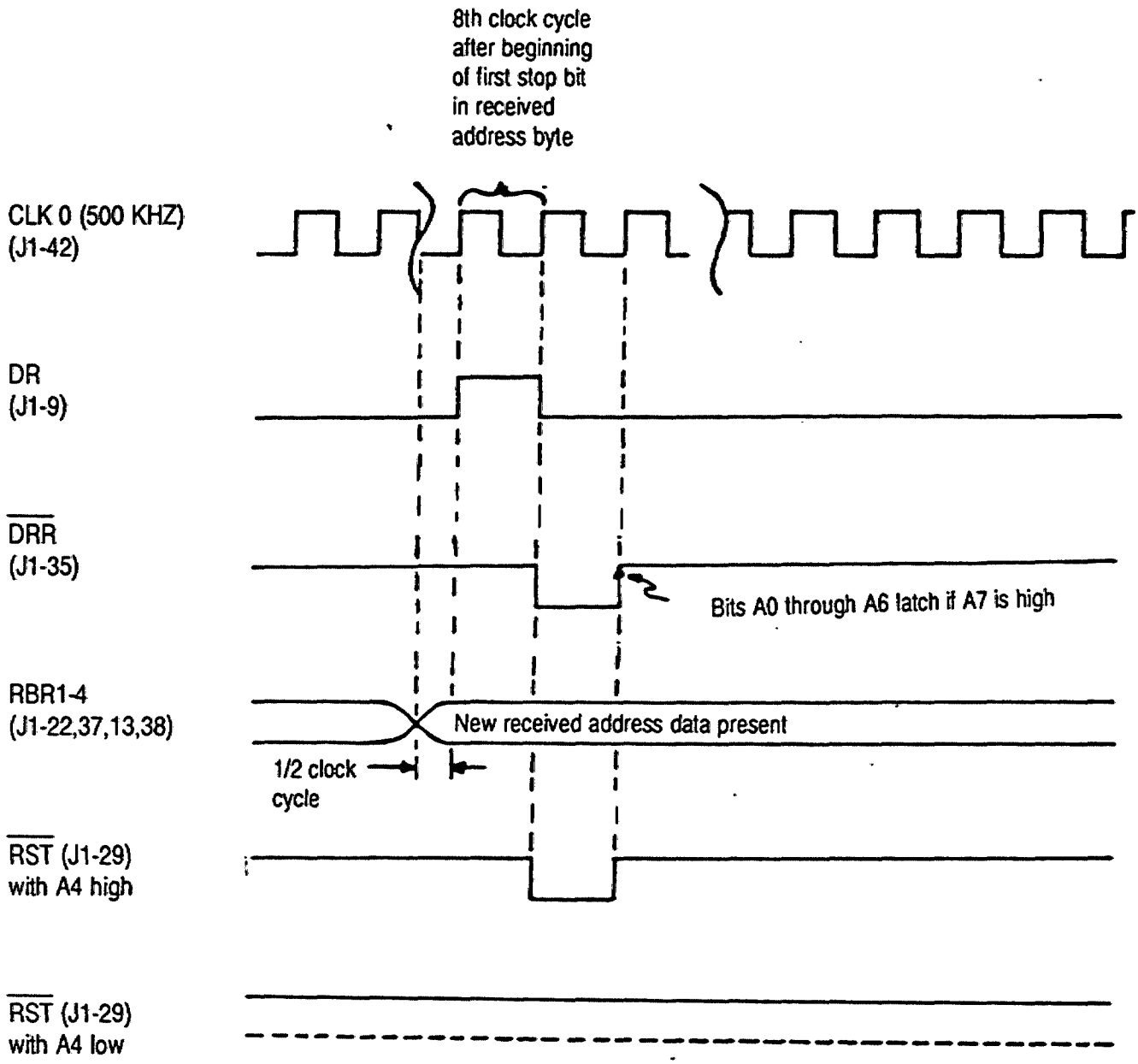


FIGURE 4. UART receiver timing diagram.

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3.5.7.3 RST output pulse. For received address bytes in which the A4 bit is high, RST (J1-29) shall pulse low once together with DRR. For received address bytes in which the A4 bit is low, RST shall remain high.

3.5.7.4 Address latching operation. For received address bytes in which the A7 bit is high, the low to high transition of DRR shall latch the A0 through A6 bits onto latch output lines S1 (J1-24), S6 (J1-50), S4 (J1-25), S3 (J1-23), SW2 (J1-48), MM0 (J1-28) and MM1 (J1-39), respectively. These bits shall remain latched until cleared by a low on MR (J1-17) or until a new address byte is received and latched. The A6 (MM1) and A5 (MM0) latched bits shall select a particular 64:8 digital multiplexer data word per Table I (Address bits for data word selection).

TABLE I. Address bits for data word selection.

A6 (MM1)	A5 (MM0)	Data Word Selected
0	0	Range and Mode
0	1	Rate and Gain
1	0	Analog Voltage and Digital Counter
1	1	Digital Test Point

3.5.8 64-line to 8-line digital multiplexer circuit. The 64:8 digital multiplexer shall consist of 8 individual 8-line to 1-line digital multiplexers. The output of each 8:1 multiplexer shall be connected to a separate UART Transmitter Buffer Register line, TBR1 (LSB), 2, 3, 4, 5, 6, 7 or 8 (MSB).

3.5.8.1 Multiplexer input addressing. The data bit inputs to the 64:8 multiplexer are digital "0's" and "1's" from TTL signals with frequencies less than 25 kHz. Unit 4BB1 address and control lines MM0, MM1 and MM2 are connected to each multiplexer chip's A, B and C address lines, respectively. With power per 3.5.2 and clock per 3.5.3 applied to the Unit 4BB1 and an address byte received per 3.5.7, MM2, controlled by the UART's Transmitter Register Empty line, shall enable the low data byte, when low, and then go high and enable the high data byte. Received and latched address bits MM0 and MM1 shall select the data word per Table II (Low byte addressing) and Table III (High byte addressing). For each combination of MM0, MM1 and MM2, the correct eight multiplexer data input lines shall be selected and

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digital "0's" and "1's" input on these lines shall appear within 50 nanoseconds on the corresponding multiplexer output lines, TBR1 through TBR8, per Tables II and III. In a weapon system, a digital "0" will carry (represent) a value of zero and a "1" will carry the weight listed beside each TBR line in Tables II and III. For Digital Counter/Analog to Digital Converter (ADC) word selection, two address bytes, not just one, shall be received per Figure 2. The first address byte shall select the Counter/ADC word and set or reset SW2, S1, S3, S4 and S6 at J1-48, 24, 23, 25 and 50, respectively. The second address byte shall cause RST (J1-29) to go logic low and set or reset RBR1, 2, 3 and 4 at J1-22, 37, 13 and 38, respectively.

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TABLE II. Low byte addressing.

Address: Byte Selector MM2 (J1-41)	Address: Word Selectors MM1(A6) (J1-39)	Address: Word Selectors MM0(A5) (J1-28)	MUX Inputs Selected	MUX Word Selected	Data Bits on Each Transmitter Buffer Register Line	MUX Data Input Connector Pin	Transmitted/ Reassembled Output Word Data Bit Location Per Figure 3
					TBR1: 1.25M	P1-9	D0
					TBR2: 2.5M	P1-13	D1
					TBR3: 5M	P1-18	D2
					TBR4: 10M	P1-22	D3
					TBR5: 20M	P1-36	D4
					TBR6: 40M	P1-41	D5
0	0	0	D0	Range & Mode	TBR7: 80M	P1-46	D6
					TBR8: 160M	P1-52	D7
					TBR1: Not Used	P1-11	D0
					TBR2: Not Used	P1-15	D1
					TBR3: 0.625M/S	P1-20	D2
0	0	0	D1	Range & Mode	TBR4: 1.25 M/S	P1-33	D3
					TBR5: 2.5 M/S	P1-38	D4
					TBR6: 5 M/S	P1-43	D5
					TBR7: 10 M/S	P1-48	D6
					TBR8: 20 M/S	P1-51	D7

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TABLE II. Low byte addressing. (continued)

Address Byte Selector Word Selectors MM2 (J1-41)	MM1(A6) (J1-39)	MM0(A5) (J1-28)	MUX Inputs Selected	MUX Word Selected	Data Bits on Each Transmitter Buffer Register Line	MUX Data Input Connector Pin	Transmitted/ Reassembled Output Word Data Bit Location Per Figure 3
					TBR1: AD1	J1-45	D0
					TBR2: AD2	J1-47	D1
					TBR3: AD3	J1-19	D2
					TBR4: AD4	J1-12	D3
					TBR5: AD5	J1-8	D4
					TBR6: AD6	J1-6	D5
0	1	0	D2	Analog and Counter	TBR7: AD7	J1-33	D6
					TBR8: AD8	J1-30	D7
					TBR1: TP1	Internal	D0
					TBR2: $\overline{200M}$	P1-17	D1
					TBR3: TP3	Internal	D2
0	1	1	D3	Test Point	TBR4: Range Cal Ind	P1-35	D3
					TBR5: $\overline{R\ RESET}$	P1-40	D4
					TBR6: C Ø	J1-31	D5
					TBR7: ECM	P1-50	D6
					TBR8: SPARE	P1-56	D7

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TABLE III. High byte addressing.

Address: Byte Selector Word Selectors MM2 (J1-41)	MM1(A6) (J1-39)	MM0(A5) (J1-28)	MUX Inputs Selected	MUX Word Selected	Data Bits on Each Transmitter Buffer Register Line	MUX Data Input Connector Pin	Transmitted/ Reassembled Output Word Data Bit Location Per Figure 3
					TBR1: 320M	P1-10	D8
					TBR2: 640M	P1-14	D9
					TBR3: 1280M	P1-19	D10
1	0	0	D4	Range & Mode	TBR4: 2560M	P1-23	D11
					TBR5: ECM	P1-37	D12
					TBR6: G3	P1-42	D13
					TBR7: TRACK	P1-47	D14
					TBR8: Not Used	P1-53	D15
					TBR1: 40 M/S	P1-12	D8
					TBR2: 80 M/S	P1-16	D9
					TBR3: 160 M/S	P1-21	D10
1	0	1	D5	Rate & Gain	TBR4: +RS	P1-34	D11
					TBR5: G1	P1-39	D12
					TBR6: G2	P1-44	D13
					TBR7: G3	P1-49	D14
					TBR8: MID RNG CAL	P1-55	D15

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TABLE III. High byte addressing. (continued)

Address Byte Selector Word Selectors MM2 (J1-41)	MM1(A6) (J1-39)	MM0(A5) (J1-28)	MUX Inputs Selected	MUX Word Selected	Data Bits on Each Transmitter Buffer Register Line	MUX Data Input Connector Pin	Transmitted/ Reassembled Output Word Data Bit Location Per Figure 3
					TBR1: C1	J1-49	D8
					TBR2: C2	J1-21	D9
					TBR3: C3	J1-46	D10
1	1	0	D6	Analog and Counter	TBR4: C4	J1-18	D11
					TBR5: C5	J1-32	D12
					TBR6: C6	J1-7	D13
					TBR7: C7	J1-27	D14
					TBR8: C8	J1-26	D15
					TBR1: TP9/SW2	J1-48	D8
					TBR2: SW1	J1-40	D9
					TBR3: SW3	J1-20	D10
1	1	1	D7	Test Point	TBR4: SW4	J1-43	D11

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3.5.8.2 Conditioned inputs to 64:8 multiplexer.

3.5.8.2.1 TP1 (7.5 MHz) input. With the following input conditions, bit D0 of the transmitted/reassembled test point word shall be high:

a. Power applied per 3.5.2.

b. 500 kHz clock applied per 3.5.3.

c. 7.5 ±3.5 MHz clock applied to P1-1.

d. Address byte with bits A5, A6, and A7 high has been received by the Unit 4BB1.

With the 7.5 ±3.5 MHz clock signal interrupted, bit D0 of the first or second and following test point words transmitted (after disconnect) shall be low.

3.5.8.2.2 TP3 (ROVF) input. With the following input conditions, a RST output pulse per 3.5.7.3 shall occur and bit D2 of the transmitted/reassembled test point word shall be high:

a. Power applied per 3.5.2.

b. 500 kHz clock applied per 3.5.3.

c. Address byte with bits A4, A5, A6, and A7 high has been received by the Unit 4BB1.

With a low-to-high transition applied to P1-8 (ROVF), bit D2 of the transmitted/reassembled test point word shall go and remain low until another RST pulse per 3.5.7.3 occurs.

3.5.9 UART transmitter circuit. Refer to timing diagram of Figure 5.

3.5.9.1 Transmitter initialization. With power and clock applied per 3.5.2 and 3.5.3, respectively, the Unit 4BB1 shall initialize (reset per 3.5.6) for transmitting data.

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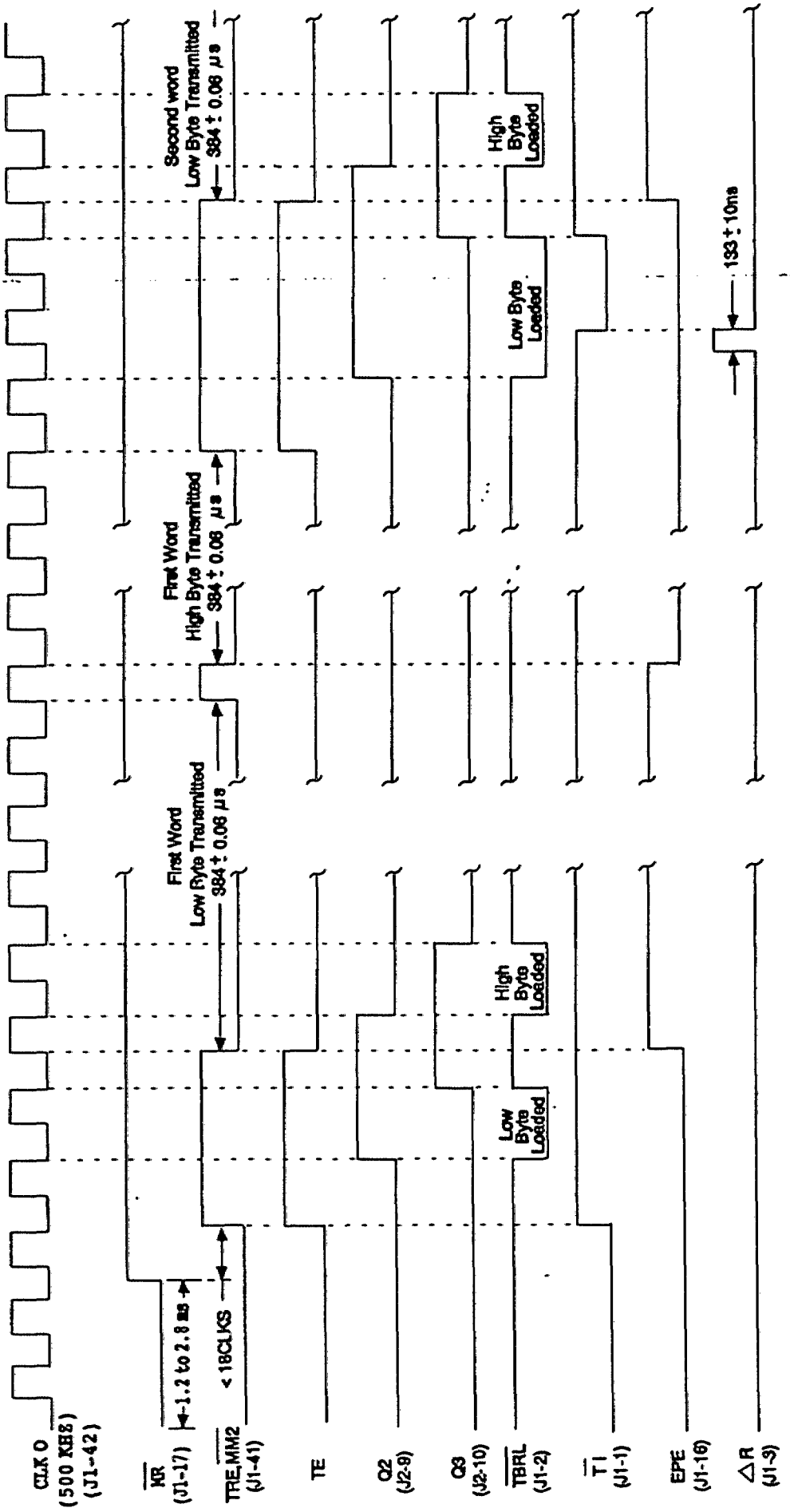


FIGURE 5. UART transmitter timing diagram.

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3.5.9.2 UART transmitter operation.

3.5.9.2.1 Multiplexer word selection. Refer to Table I.

a. When no address bytes have been received, the Range and Mode word shall be selected, because a master reset per 3.5.6 shall clear MM1 and MM0 to zero.

b. Each word shall be selectable by inputting an address byte with latch bit A7 high and bits A6(mm1) and A5(MM0) set to the binary code for the desired word.

3.5.9.2.2 UART loading and parity generation.

a. The Johnson Counter (ring counter) shall be continuously cycled by the 500 kHz clock and cause two UART load ($\overline{\text{TBRL}}$ low) signals each time the transmitter is enabled (TE goes high).

b. After master reset, MM2 and EPE shall be low, therefore, when the first UART load signal occurs, the low data byte shall be loaded from TBR1 through TBR8 and assigned odd parity.

c. TRE shall transition from high to low, when the low data byte is transferred from the UART's Transmitter Buffer Register to the UART's Transmitter Register, and cause MM2 and EPE to go high.

d. With MM2 and EPE high, when the second load signal occurs, the high data byte shall be loaded from TBR1 through TBR8 and assigned even parity.

e. After the low byte is transmitted out, TRE shall go high. TRE shall then return low when the Transmitter Register receives the high byte of the first word. The high-to-low transition of TRE shall cause MM2 and EPE to return low to get the low byte of the next word and assign it odd parity.

f. When an ΔR pulse (range is changing signal) comes into the Unit 4BB1, while the UART is loading ($\overline{\text{TBRL}}$ low), the trailing edge of that ΔR pulse shall cause the Johnson Counter to inhibit counting for 1 to 2 clock cycles.

g. Inhibiting counting shall cause $\overline{\text{TBRL}}$ to remain low longer to allow the UART to load the new nonchanging range data, and prevent the transmission of disjointed changing data.

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3.5.9.2.3 UART data transmission.

a. Every low-to-high transition of $\overline{\text{TBRL}}$ generated by the Johnson Counter, shall transfer a data byte from the UART's Transmitter Buffer Register to the UART's Transmitter Register.

b. Each data byte in the Transmitter Register shall be serially clocked out TRO (J1-34) and UNIT 4 to CPU DATA (P1-6 HI, P1-7 LO) with format per 3.5.5 and complete with start, parity and two stop bits. Transmitter timing shall be per Figure 5.

c. Each multiplexer input bit shall appear at its appropriate data location in the transmitted and reassembled output data word per Tables II and III.

3.5.10 Input/output requirements. Unit 4BB1 inputs shall be as listed in Tables IV and V. Unit 4BB1 outputs shall be as listed in Tables VI and VII. These tables contain low logic input voltages (LLIVs), high logic input voltages (HLIVs), low logic output voltages (LLOVs), high logic output voltages (HLOVs), low logic output currents (LLOIs) and high logic output currents (HLOIs).

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TABLE IV. P1 inputs.

Connector Pin	Signal Name	MAX LLIV	MIN HLIV	Descriptive Data	General Range
1	7.5 MHZ CLK	.7V	2V	7.5 MHZ	0-5V
2	UNIT 4 CLOCK (500 KHZ) HI	.8V	2V	500 KHZ DiFF.INPUT	0-5V
3	UNIT 4 CLOCK (500 KHZ) LO	.8V	2V	CLOCK	0-5V
4	CPU TO UNIT 4 DATA HI	.8V	2V	SERIAL INPUT	0-5V
5	CPU TO UNIT 4	.8V	2V	DATA DIFF.	0-5V
8	ROVF	.7V	2V		0-5V
9	1.25 M	.7V	2V		0-5V
10	320 M	.7V	2V		0-5V
11	SPARE INPUT	.7V	2V		0-5V
12	40 M/S	.7V	2V		0-5V
13	2.5 M	.7V	2V		0-5V
14	640 M	.7V	2V		0-5V
15	SPARE INPUT	.7V	2V		0-5V
16	80 M/S	.7V	2V		0-5V
17	$\overline{200\ M}$	.7V	2V		0-5V
18	5 M	.7V	2V		0-5
19	1280 M	.7V	2V		0-5V
20	0.625 M/S	.7V	2V		0-5V
21	160 M/S	.7V	2V		0-5V
22	10 M	.7V	2V		0-5V
23	2560 M	.7V	2V		0-5V

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TABLE IV. P1 inputs. (continued)

Connector Pin	Signal Name	MAX LLIV	MIN HLIV	Descriptive Data	General Range
33	1.25 M/S	.7V	2V		0-5V
34	+RS	.7V	2V		0-5V
35	RANGE CAL IND	.7V	2V		0-5V
36	20 M	.7V	2V		0-5V
37	ECM	.7V	2V		0-5V
38	2.5 M/S	.7V	2V		0-5V
39	G1	.7V	2V		0-5V
40	$\overline{\text{R RESET}}$	.7V	2V		0-5V
41	40 M	.7V	2V		0-5V
42	G3	.7V	2V		0-5V
43	5 M/S	.7V	2V		0-5V
44	G2	.7V	2V		0-5V
45	SPARE INPUT	.7V	2V		0-5V
46	80 M	.7V	2V		0-5V
47	TRACK	.7V	2V		0-5V
48	10 M/S	.7V	2V		0-5V
49	G3	.7V	2V		0-5V
50	ECM	.7V	2V		0-5V
52	160 M	.7V	2V		0-5V
53	SPARE INPUT	.7V	2V		0-5V
54	20 M/S	.7V	2V		0-5V
55	MID RANGE CAL	.7V	2V		0-5V
56	SPARE INPUT	.7V	2V		0-5V

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TABLE V. J1 inputs.

Connector Pin	Signal Name	MAX LLIV	MIN HLIV	Descriptive Data	General Range
3	R	.7V	2V		0-5V
4	+5 VDC			POWER INPUT	
5	+5 VDC			POWER INPUT	
6	AD6	.7V	2V		0-5V
7	C6	.7V	2V		0-5V
8	AD5	.7V	2V		0-5V
10	SW6	.7V	2V		0-5V
12	AD4	.7V	2V		0-5V
14	+5 VDC RET			POWER RETURN	0-5V
15	+5 VDC RET			POWER RETURN	0-5V
18	C4	.7V	2V		0-5V
19	AD3	.7V	2V		0-5V
20	SW3	.7V	2V		0-5V
21	C2	.7V	2V		0-5V
26	C8	.7V	2V		0-5V
27	C7	.7V	2V		0-5V
30	ADS	.7V	2V		0-5V
31	C0	.7V	2V		0-5V
32	C5	.7V	2V		0-5V
33	AD7	.7V	2V		0-5V
40	SW1	.7V	2V		0-5V
43	SW4	.7V	2V		0-5V
45	AD1	.7V	2V		0-5V
46	C3	.7V	2V		0-5V
47	AD2	.7V	2V		0-5V
49	C1	.7V	2V		0-5V

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TABLE VI. P1 outputs.

Connector Pin	Signal Name	MAX LLOV	MIN HLOV	MAX LLOI	MAX HLOI	Descrip-tive Data	Gen-eral Range
6	UNIT 4 TO CPU DATA HI	.5V	2.5V	20MA	-20MA	DIFF. SERIAL	0-5V
7	UNIT 4 TO CPU DATA LO	.5V	2.5V	20MA	-20MA	DATA OUTPUT	0-5V

TABLE VII. J1 outputs.

Connector Pin	Signal Name	MAX LLOV	MIN HLOV	MAX LLOI	MAX HLOI	General Range
13	RBR3	.45V	2.4V	2MA	-.2MA	0-5V
22	RBR1	.45V	2.4V	2MA	-.2MA	0-5V
23	S3	.4V	2.5V	4MA	-400UA	0-5V
24	S1	.4V	2.5V	4MA	-400UA	0-5V
25	S4	.4V	2.5V	4MA	-400UA	0-5V
29	$\overline{\text{RST}}$	.4V	2.5V	4MA	-400UA	0-5V
37	RBR2	.45V	2.4V	2MA	-.2MA	0-5V
38	RBR4	.45V	2.4V	2MA	-.2MA	0-5V
42	CLK 0 (500 KHZ)	.45V	2.5V	8MA	-400UA	0-5V
48	SW2	.4V	2.5V	4MA	-400UA	0-5V
50	S6	.4V	2.5V	4MA	-400UA	0-5V

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3.5.11 Test points. The Unit 4BB1 shall contain test points as listed in Table VIII.

TABLE VIII. Test points.

Connector Pin	Signal Name	MAX LLOV	MIN HLOV	MAX LLOI	MAX HLOI	Descriptive Data	General Range
J1-1	$\overline{TI}$	.45V	2.5V	8MA	-440UA		0-5V
J1-2	$\overline{TBRL}$	.4V	2.5V	4MA	-440UA		0-5V
J1-9	DR	.45V	2.4V	2MA	.2MA	DATA REC'D	0-5V
J1-16	EPE	.4V	2.5V	4MA	-400UA		0-5V
J1-17	$\overline{MR}$	.4V	2.5V	4MA	-400UA		0-5V
J1-28	MMO	.4V	2.5V	4MA	-400UA		0-5V
J1-34	TRO	.45V	2.4V	2MA	-.2MA	TRANS REG OUTPUT	0-5V
J1-35	$\overline{DRR}$	.4V	2.5V	4MA	-400UA		0-5V
J1-36	RRI	.45V	2.5V	8MA	-400UA	INPUT SERIAL DATA	0-5V
J1-39	MM1	.4V	2.5V	4MA	-400UA		0-5V
J1-41	MM2	.4V	2.5V	4MA	-400UA		0-5V
J1-44	Q1	.4V	2.5V	4MA	-400UA		0-5V
J2-1	TBR8	.4V	2.5V	4MA	-400UA		0-5V
J2-2	TBR7	.4V	2.5V	4MA	-400UA		0-5V
J2-3	TBR6	.4V	2.5V	4MA	-400UA		0-5V
J2-4	TBR5	.4V	2.5V	4MA	-400UA		0-5V

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TABLE VIII. Test points. (continued)

Connector Pin	Signal Name	MAX LLOV	MIN HLOV	MAX LLOI	MAX HLOI	Descrip-tive Data	Gen-eral Range
J2-5	TBR4	.4V	2.5V	4MA	-400UA		0-5V
J2-6	TBR3	.4V	2.5V	4MA	-400UA		0-5V
J2-8	TBR1	.4V	2.5V	4MA	-400UA		0-5V
J2-9	Q2	.4V	2.5V	4MA	-400UA		0-5V
J2-10	MR	.4V	2.5V	4MA	-400UA		0-5V

3.5.12 Spare connector pins. The Unit 4BB1 shall contain spare pins as listed in Table IX.

TABLE IX. Spare pins.

CONNECTOR-PIN
J1-11
P1-24, 32, 51

3.6 Soldering. All soldering shall be in accordance with the requirements of Note 2 of Drawing 12950877.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspections set forth in this specification shall become part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Special sampling inspection (see 4.5).

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4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+60°F to +90°F (+16°C to +32°C)
b.	Relative humidity	90% RH max.
c.	Pressure	28 to 32 inches of Hg (700 to 815 mmHg)

4.3 First article inspection.

4.3.1 Sample. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the assemblies, components and test specimens listed below in the quantities indicated.

Name	Drawing	Quantity
Unit 4BB1	9360949	3

4.3.2 Inspections to be performed: As determined by the Government, the first article assemblies, components and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification and be inspected for compliance with any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component, or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of any assembly, component or test specimen to comply with any of the requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to

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exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies.

a. The contractor's quality program or detailed inspection plan shall provide assurance of compliance of all characteristics with the applicable drawings and specifications utilizing as a minimum the conformance criteria specified in this document and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

b. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, other types of sampling plans, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.3). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality conformance provision(s) herein. In case of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The

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tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot.

4.4.2.3 Functional tests. The tests in Table X shall be performed on a 100% basis.

TABLE X. Functional tests (100 percent inspection).

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
101. Electrical	3.4	4.7.2
102. Performance	3.5	4.7.3
Minor:	None	N/A

4.5 Special sampling. One Unit 4BB1 out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests of Table X and shall then meet the tests in Table XI. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

TABLE XI. Special sampling.

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
103. Vibration	3.3.1	4.7.1.1
104. Shock	3.3.2	4.7.1.2
105. Altitude	3.3.3	4.7.1.3
106. Temperature	3.3.4	4.7.1.4
107. Humidity	3.3.5	4.7.1.5
Minor:	None	N/A

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4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

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4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table XII (Test equipment list) shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

TABLE XII. Test equipment list.

ITEM NUMBER	DESCRIPTION
1. Interface Device C	9397373 with PWB W5 (9397562) W4 Cable (9397563), and test cable W6 (9397376-2)
2. Digital Card Tester	GenRad 2225A (AN/USM-465A)
3. ComCode Program Tape	Version 2.0
4. Test Program Set Manual	TPS 9361338
5. Test Program Tape	TPT 9391338
6. Oscilloscope	Input impedance of 1 Mohm or greater, single sweep capability, with probe

4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810, except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514, Category 8, Procedure I, of MIL-STD-810 and paragraph 3.3.1 herein. In Step 1 of Procedure I only steps 4 through 8 of paragraph II-2.1 apply. The Unit 4BB1 shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the Unit 4BB1 shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

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4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 hertz.

After the application of all the shock impulses, the Unit 4BB1 shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (operating). This test shall be performed as specified in Procedures II, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and nonoperating). This test shall be performed as specified in Procedures I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.4.2 Low (operating and nonoperating). This test shall be performed as specified in Procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Humidity (nonoperating). This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment and visual examination for compliance with 3.4.

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4.7.3 Performance.

4.7.3.1 Master reset circuit. The master reset circuit shall be tested to meet the requirements of 3.5.6.

4.7.3.2 UART receiver circuit. The UART receiver circuit shall be tested to meet the requirements of 3.5.7.

4.7.3.3 Digital multiplexer circuit. The digital multiplexer circuit shall be tested to meet the requirements of 3.5.8.

4.7.3.4 UART transmitter circuit. The UART transmitter circuit shall be tested to meet the requirements of 3.5.9.

4.8 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of Note 2 of Drawing 12950877 as specified in paragraph 3.6.

5. PACKAGING

5.1 General. Preservation, packing and marking shall be in accordance with Special Packaging Instruction PS 9360884. The level of protection shall be as specified in the procurement documents.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Unit 4BB1 shall be used in the Radar Computer Box of the VADS self-propelled and towed vehicles.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

c. Applicable stock number.

d. First article sample requirements.

e. Packaging requirements, if other than specified in Section 5.

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f. Serialization requirements, if applicable.

g. Certification of conformance for each lot of shipment of product.

6.3 Submission by the contractor.

a) Inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army ARDED, ATTN: SMCAR-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

b) Alternative quality conformance provisions for approval. Unless otherwise specified in the contract, proposed alternative quality conformance provisions will be submitted by the contractor for evaluation to the technical activity responsible for the preparation of this specification.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Subject term (key word) listing.

Fire Control System (FCS)
Radar Set AN/VPS-2A
Vulcan Air Defense System (VADS)

6.6 Changes from previous issue. The previous issues were for ARDEC internal use only. They were not forwarded to Naval Publications and Forms Center (NPFC) for publication. This issue is initial public release for unlimited distribution.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1285-A 887)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-70571	23 FEBRUARY 1993

3. DOCUMENT TITLE: CIRCUIT BOARD, BUILT-IN-TEST BOARD NO. 1, UNIT 4: 9360949

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial (201) 724-6674	(2) AUTOVON 880-6674
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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